Duplicate leakage in a public conjunctival pallor benchmark, and an honest baseline for image-based anaemia screening

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Abstract

Conjunctival pallor photographs are an attractive substrate for non-invasive anaemia screening, and CP-AnemiC — 710 images of Ghanaian children aged 6–59 months with paired laboratory haemoglobin — has become a convenient public benchmark for the task. We report that the benchmark contains duplication substantial enough to change published conclusions, and we provide a corrected baseline.

Only 383 of the 710 metadata rows are distinct photographs. The duplication takes three forms, each of which defeats the detection method that sufficed for the previous one: 212 byte-identical copies; 19 re-encoded copies whose bytes, and so whose hashes, differ; and 96 *shifted or re-cropped* copies whose hand-drawn masks differ as well, so that neither hashing nor bounding-box pixel comparison detects them. 116 groups carry *conflicting* haemoglobin labels on the same photograph, one spanning 3.3 to 10.4 g/dL. Because copies are frequently attributed to different hospitals, the redundancy defeats site-grouped cross-validation as well as random splits: under hash and aligned-pixel deduplication with site-grouped folds, 74 photographs still enter the analysis as 164 rows across different site folds. A random split inflates AUROC by +0.176 (95 % CI +0.120 to +0.232, $n_{\text{boot}} = 5000$); a matched-n control attributes about half of the final deduplication step to leakage and about half to the smaller resulting sample.

After removing all three forms and grouping by collection site, twelve interpretable colour features reach AUROC 0.706 (95 % CI 0.653–0.757) for WHO anaemia (Hb $<$ 11 g/dL), against a demographics-only floor of 0.524; the image contributes +0.182 AUROC (DeLong 95 % CI +0.107 to +0.258, $p = 2.1 \times 10^{-6}$). Bland–Altman limits of agreement span 7.8 g/dL with strong proportional bias (slope −0.81), so the model is a triage screen and not a haemoglobin meter. At 90 % sensitivity, specificity falls to 0.303, and decision curve analysis places the model below referring every child up to a threshold probability of about 0.26 — it is useful only where confirmatory testing is scarce.

We recommend that future work on CP-AnemiC deduplicate under small translations, not merely by file hash or aligned pixel comparison, and we release the detector and full analysis so that reported numbers can be checked against these controls.

Keywords: anaemia screening; conjunctival pallor; data leakage; benchmark integrity; medical imaging; low-resource diagnostics.

1 Introduction

Anaemia affects an estimated 40 % of children aged 6–59 months worldwide, rising to 68 % in west and central Africa [2], and diagnosis conventionally requires a blood draw and laboratory or point-of-care haemoglobin assay. Both are constrained in low-resource settings by cost, consumables, trained staff and infection-control requirements. This has motivated interest in image-based screening: the palpebral conjunctiva is a mucous membrane whose colour is visibly influenced by haemoglobin concentration, is straightforward to expose, and can be photographed with a consumer smartphone.

Public datasets are what allow such methods to be compared. CP-AnemiC [1] pairs conjunctiva photographs of Ghanaian children with laboratory haemoglobin and has been adopted as a benchmark for this task. The value of any benchmark, however, rests on an assumption that is rarely re-examined once a dataset is in circulation: that its records are distinct. Where they are not, a randomly partitioned evaluation places copies of the same photograph in both training and test folds, and the resulting score measures memorisation rather than generalisation.

This paper makes three contributions.

1. We show that CP-AnemiC contains duplicate records in three forms — byte-identical, re-encoded, and shifted or re-cropped — at a scale that materially inflates reported performance, and we quantify that inflation. The third form is detectable only under translation alignment, and because its members are frequently attributed to different hospitals it defeats site-grouped validation as well as random splits.

2. We establish a corrected baseline under full deduplication and site-grouped cross-validation, using interpretable colour features only.

3. We characterise the resulting model as a screening instrument — reporting the operating point, calibration and Bland–Altman agreement that determine whether it could be deployed — and conclude that it is not yet adequate as a haemoglobin estimator.

We propose no new architecture; the aim is to establish what this benchmark supports under stated controls, and to make those controls reproducible.

2 Data

2.1 Source

CP-AnemiC [1] comprises conjunctiva photographs of children aged 6–59 months collected at ten hospitals across four Ghanaian regions, each paired with a laboratory haemoglobin measurement. Ages in the distributed spreadsheet are bucketed (11, 12, 24, 36, 48 and 60 months), and 90 rows are coded at 60 months, marginally outside the stated band; we use age only as a demographic covariate and as the demographics-only baseline. We use the WHO threshold for childhood anaemia, Hb $<$ 11 g/dL [3]. Table 1 summarises the dataset as distributed and after the deduplication described below.

2.2 Duplicate structure

Three distinct forms of redundancy are present, and each is invisible to the method that suffices for the one before it.

Byte-identical duplication accounts for 212 files: the same image registered under more than one identifier, detectable by hashing the file (710 $\rightarrow$ 498).

Re-encoded copies account for a further 19: the same photograph written to disk again, so the bytes and therefore the hash differ. Thirteen are exactly pixel-identical once the region of

Table 1: CP-AnemiC as distributed and after deduplication.

Property	Value
Metadata rows (as distributed)	710
Distinct files by SHA-256 (pass 1)	498
Distinct after re-encode merge (pass 2)	479
Distinct photographs after alignment (pass 3)	**383**
Rows whose file hash appears exactly once	407
Duplicate groups with conflicting Hb (hash grain)	90
the same, at the 383-photograph grain	116
Groups attributed to more than one hospital	125
Collection sites	10 hospitals, 4 regions
Cohort	children 6–59 months
Label	laboratory Hb (g/dL); WHO threshold 11
Anaemia prevalence after dedup	0.509

interest is extracted and six differ only by a small crop or in under 2 % of pixels. Detecting them requires comparing decoded pixels within the annotated region (498 → 479).

Shifted or re-cropped copies are the largest group, at 96, and the least visible. Here the photograph is the same but the hand-drawn mask is not, so the region of interest is translated or trimmed relative to its twin. Both the file hash and the bounding-box-normalised pixel comparison fail: the bytes differ, and the extracted regions no longer align. Detecting them requires searching over small translations at full resolution (479 → 383). Figure 1 shows one such pair.

The consequential subsets are two. At the final grain, 116 groups carry conflicting haemoglobin labels on the same photograph — one spanning 3.3 to 10.4 g/dL. Because the pixels are the same, no model can satisfy both labels; the disagreement is irreducible label noise that places a ceiling on achievable performance. Where duplicates were merged we took the median haemoglobin, which is a defensible summary but cannot recover the truth. Separately, 125 groups are attributed to more than one hospital, and this is the more damaging property for evaluation. Grouping folds by collection site — the standard remedy for site-specific confounds — cannot keep copies together when the copies disagree about which site they came from. Concretely, under a protocol that deduplicates by hash and by aligned pixels and then groups by site, 74 photographs still enter the analysis as 164 separate rows assigned to more than one site fold; under hash-only deduplication, 83 do. A protocol we would previously have called correct still leaks.

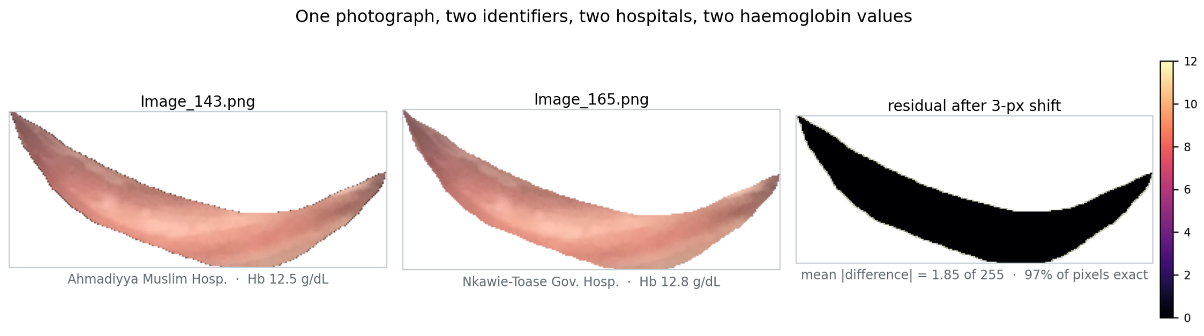

Figure 1: A shifted duplicate pair. The two files have different SHA-256 hashes, different hand-drawn masks, and are attributed to different hospitals with different haemoglobin values. After a three-pixel vertical shift, 97 % of overlapping pixels are exactly equal and the mean absolute difference is 1.85 of 255 — a residual consistent with re-compression, not with two photographs. Neither file hashing nor bounding-box pixel comparison detects this pair.

2.3 Mask handling

Each image is RGBA, where the alpha channel is a hand-drawn conjunctiva mask covering approximately 25 % of the frame; the remainder is black padding. Colour statistics must therefore be computed over masked pixels only. Ignoring the mask shifts mean RGB by more than 100/255 and produces features that encode *mask area* — an artefact of annotator behaviour — rather than tissue colour.

3 Methods

3.1 Features

We extract twelve interpretable colour features over the masked region of interest: mean R, G, B; circular mean hue and hue concentration; mean saturation; CIELAB L^*, a^*, b^*; an illumination-normalised redness index $r/(r + g + b)$; its 10th percentile; and the red/green ratio. The a^* channel is the perceptual red–green axis that corresponds to what a clinician reads as pallor.

Two exclusions are deliberate. **Mask area is not used as a feature**: it reflects how the annotator drew the boundary rather than physiology, and had it correlated with site or severity the model would have learned the annotator. **Hue is averaged circularly**: red tissue sits near both 0.0 and 1.0 on the hue circle, so a linear mean of 0.98 and 0.02 returns 0.5, which is cyan.

3.2 Model and validation

A gradient-boosted regressor predicts haemoglobin; negating its output yields a screening score. Hyperparameters were fixed *a priori* and never tuned on this data; a nested cross-validation check (Section 4.10) confirms they were not selected for flattery: tuning inside each fold scores lower, not higher. All reported metrics are out-of-fold.

Deduplication. Three passes run in sequence, each addressing what the previous one cannot see. Pass 1 hashes file bytes with SHA-256. Pass 2 decodes each remaining image, crops to the conjunctiva mask's bounding box, downsamples to 32×32 and merges pairs whose mean absolute difference falls below 3.0 on the 0–255 scale; merged pairs all lie below 2.1 while the closest distinct pair sits at 4.1, so the threshold occupies a measured gap. Pass 3 targets copies whose masks differ, which defeats the bounding-box normalisation of pass 2: candidate pairs — taken from the union of a generous thumbnail cut-off and a translation-invariant colour histogram, so that a mask mismatch cannot hide a pair from both — are verified at full resolution by the smallest masked mean absolute difference over integer translations up to ± 30 px. Same-capture pairs sit below 2.7; visually distinct repeat captures begin at 3.1, and we place the threshold at 3.0 within that gap after inspecting the borderline pairs by eye. All merges use union–find, so a chain of copies collapses to one photograph.

Cross-validation. Folds are grouped by collection site, so that a model cannot exploit site-specific camera, lighting or operator signatures shared between training and test folds. We report a waterfall over both requirements. Neither requirement alone is sufficient here. After hashing and the re-encode merge, 74 photographs still appear as multiple rows attributed to different hospitals, so site-grouping cannot contain them; and because those rows differ in bytes and in mask, hashing and bounding-box comparison do not remove them either. Where a merged group spans several sites we assign it the site of its first-listed member, a convention rather than a fact: 125 of the 383 groups have no unambiguous site.

4 Results

4.1 Progressive removal of optimism

Table 2 evaluates the same colour features under successively stricter protocols. Each row closes one more leak than the row above.

Table 2: Effect of evaluation protocol on apparent performance. Colour features only.

Configuration	n	AUROC [95 % CI]	MAE (g/dL)
Random split, duplicates kept *(common)*	710	0.883 [0.858–0.906]	1.46
Byte-identical removed, random split	498	0.796 [0.758–0.834]	1.44
+ grouped by collection site	498	0.809 [0.772–0.847]	1.42
+ re-encoded copies removed	479	0.780 [0.737–0.818]	1.46
+ shifted/re-cropped copies removed (headline)	**383**	**0.706 [0.653–0.757]**	**1.60**

The naive configuration — a random split with duplicates retained, which is the default when a dataset is assumed clean — reports AUROC 0.883. The corrected protocol reports 0.706. A bootstrap over 5000 resamples places the total inflation at +0.176 (95 % CI +0.120 to +0.232; the interval excludes zero). That interval and the DeLong intervals in Table 3 are computed on overlapping data and are not independent tests. Figure 2 shows the waterfall.

The final step merits separate attention. Removing shifted and re-cropped copies costs a further 0.073 AUROC. That step does two things at once, however — it removes leakage *and* it removes 96 rows — so the drop must be decomposed before it is credited to leakage. Subsampling the 479-photograph set down to 383 rows at random, leaving its leakage intact, gives 0.744±0.021 over 25 draws. Roughly half of the 0.073 is therefore attributable to the smaller sample (0.036) and roughly half to the leakage the pass removes (0.038). We report both, because a step that deletes data will always depress a learning-curve-limited estimate, and attributing the whole of it to leakage would overstate the case.

An analysis that hashed files and grouped folds by site would stop at 0.780.

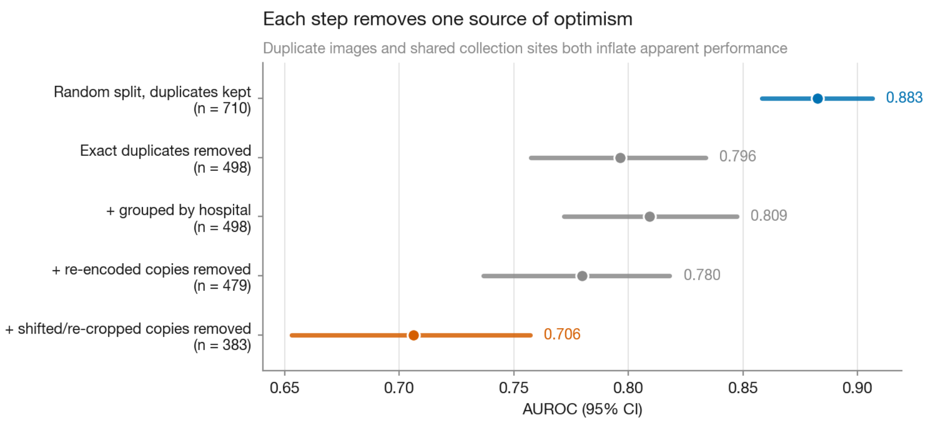

Figure 2: Apparent performance under successively stricter protocols. The fall is not uniform: removing byte-identical duplicates costs 0.087, grouping by hospital then *raises* the estimate by 0.013 (site grouping removes a confound but also changes the fold composition), and the two content-based passes cost 0.029 and 0.073. The gap between the first and last bars is the optimism introduced by evaluating on a benchmark whose records are not distinct.

4.2 Does the image contribute?

A screening score could in principle be driven by demographics alone. Table 3 rules this out.

Table 3: Feature ablation under the corrected protocol ($n = 383$). Comparisons by DeLong's test [5].

Feature set	AUROC [95 % CI]	vs. colour
Demographics only (age, sex)	0.524 [0.468–0.582]	$\Delta = -0.182$, $p = 2.1 \times 10^{-6}$
Colour only	**0.706 [0.653–0.757]**	—
Colour + demographics	0.699 [0.647–0.751]	$\Delta = -0.008$, $p = 0.55$

The image carries substantial and highly significant signal over demographics, and the demographics-only floor is now close to chance (0.524, CI spanning 0.5) — the earlier, higher floor was itself partly a duplicate artefact. Adding demographics to colour produces no detectable improvement, the expected result if the photograph already encodes what age and sex would otherwise proxy. The central claim of the paper survives the full deduplication: whatever else the third pass removed, it did not remove the image signal. Figure 3 shows the corresponding ROC curves.

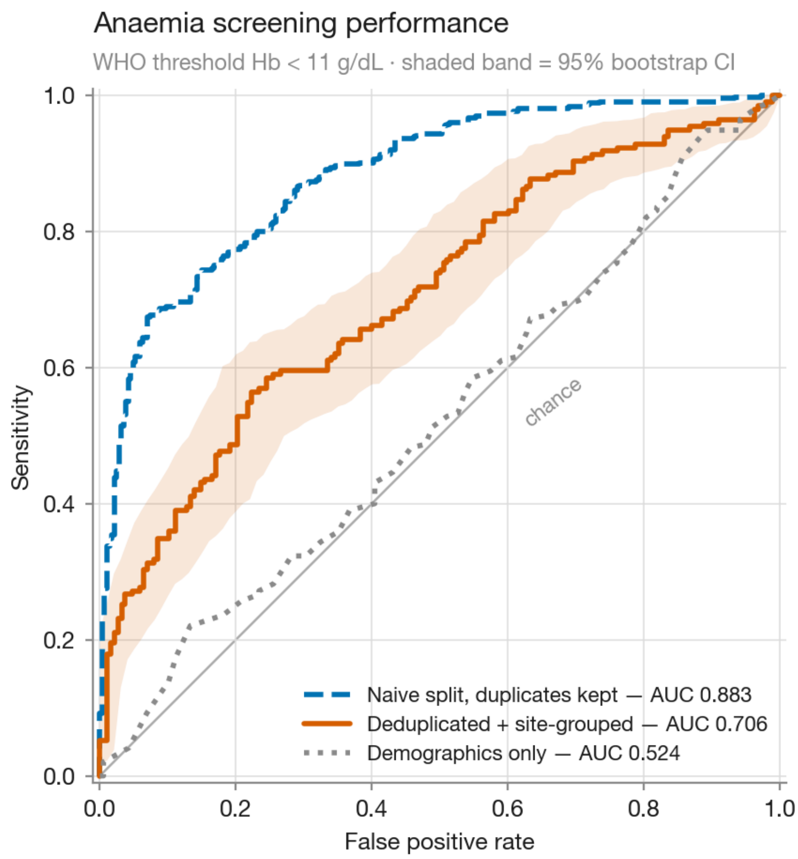

Figure 3: ROC curves. The corrected protocol (0.706) sits well below the naive leaky split (0.883) and well above the demographics-only floor (0.524). The colour + demographics variant is omitted from the plot; it is indistinguishable from colour alone (Table 3).

4.3 Choice of model

We report both a haemoglobin regressor and a direct binary classifier (Table 4), because they answer different questions and we did not wish to adopt the better number after the fact.

The classifier discriminates better, as expected when the decision is optimised directly, but it produces no haemoglobin estimate and therefore cannot be examined for clinical agreement. We retain the regressor as the headline because the agreement result (Section 4.6) is the more

Table 4: Regressor versus direct classifier.

Model	AUROC [95 % CI]	Hb output	Agreement assessable
Hb regressor *(headline)*	0.706 [0.653–0.757]	yes, MAE 1.60 g/dL	yes
Binary classifier	**0.733 [0.680–0.785]**	no	no

consequential finding.

4.4 Calibration

Table 5: Post-hoc calibration improves calibration at a cost in discrimination. Isotonic calibration applied via `CalibratedClassifierCV` with $cv = 3$.

Variant	Brier	ECE	H–L p	AUROC	prob. range
Uncalibrated	0.220	0.114	< 0.001	**0.733**	[0.01, 0.98]
Isotonic	0.220	**0.079**	**0.156**	0.711	[0.10, 0.84]

Neither variant yields probabilities fit to report. The uncalibrated model discriminates better but is clearly miscalibrated (Hosmer–Lemeshow $p < 0.001$, ECE 0.114); isotonic calibration repairs the calibration — Hosmer–Lemeshow rises to 0.156 and ECE falls to 0.079 — while costing 0.022 AUROC and compressing the probability range to $[0.10, 0.84]$. The compression is the mechanism: `CalibratedClassifierCV` fits k sub-models on $k-1$ folds each and averages them, and at roughly 300 training rows per fold each sub-model is data-starved, so averaging pulls estimates toward the centre.

We report the uncalibrated model as the headline because the instrument is a ranking triage screen, for which discrimination is the operative property, and we make the trade explicit rather than choosing the flattering column. This model can order children by risk, but its output should not be read as a probability of anaemia. We note that this section previously reported the opposite conclusion — that calibration degraded both properties — on the incompletely deduplicated data; the reversal is a reminder that model-selection findings inherit whatever contamination the evaluation set carries.

4.5 Screening operating point

At the WHO threshold the model achieves sensitivity 0.713 and specificity 0.532 (Figure 5). Tuned to catch 90 % of anaemic children, specificity falls to 0.303 — referring roughly 70 % of healthy children for confirmatory testing. At this cohort's prevalence that is about 80 referrals per 100 children screened, against 100 for referring everyone: the screen saves roughly a fifth of confirmatory tests while missing one anaemic child in ten. This, rather than AUROC, is the quantity that determines deployability, and at present it is not good enough for unsupervised use.

4.6 Agreement as a haemoglobin estimator

Bland–Altman analysis [6] gives a bias of +0.03 g/dL with limits of agreement from −3.87 to +3.92 g/dL, a span of 7.8 g/dL, and a proportional-bias slope of −0.81. The model shrinks toward the population mean (Figure 6), over-predicting in anaemic children and under-predicting in healthy ones; Figure 7 shows the agreement plot itself. A clinically useful non-invasive haemoglobin meter would require limits nearer ±1 g/dL. **No haemoglobin value from this model should be reported to a clinician.**

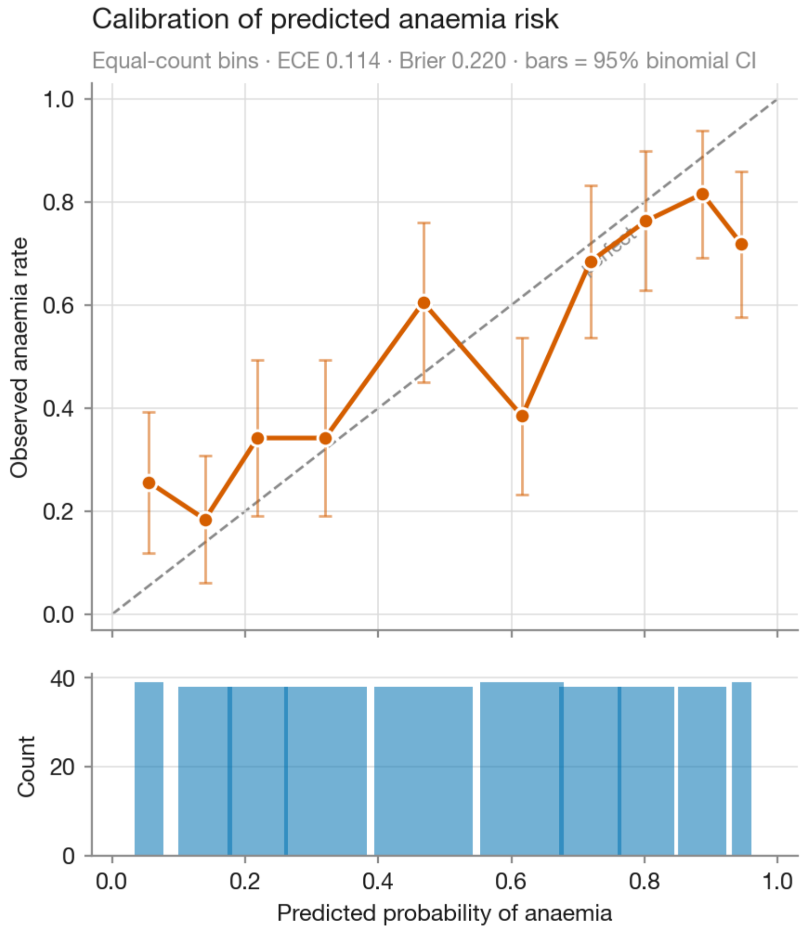

Figure 4: Reliability curve for the uncalibrated model, with equal-count bins and binomial intervals. The isotonic variant is not plotted; its summary statistics are in Table 5.

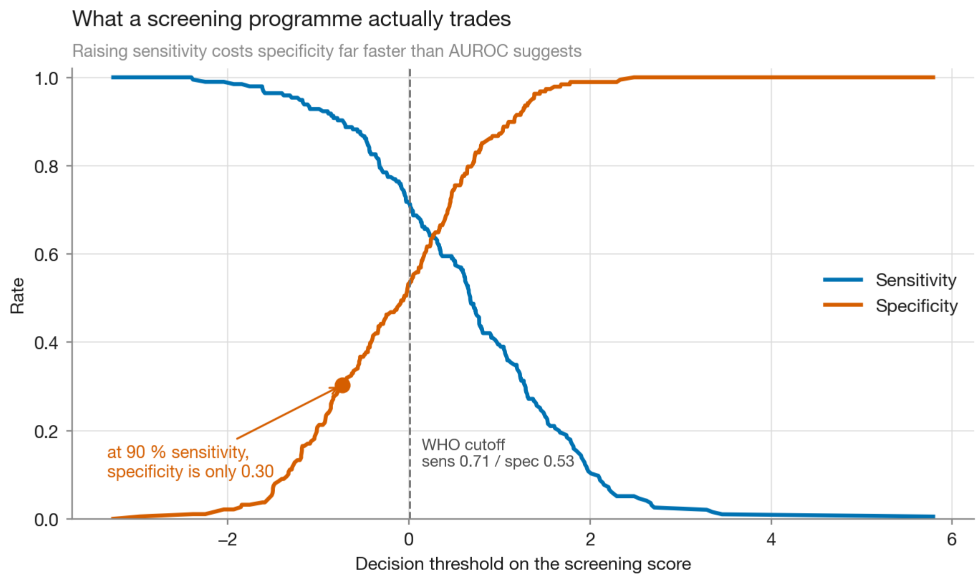

Figure 5: Sensitivity and specificity against the decision threshold, from the same out-of-fold predictions as the headline result. The curves cross steeply: moving from the WHO cutoff to 90 % sensitivity costs roughly half the specificity. This trade, rather than AUROC, is what a screening programme has to absorb.

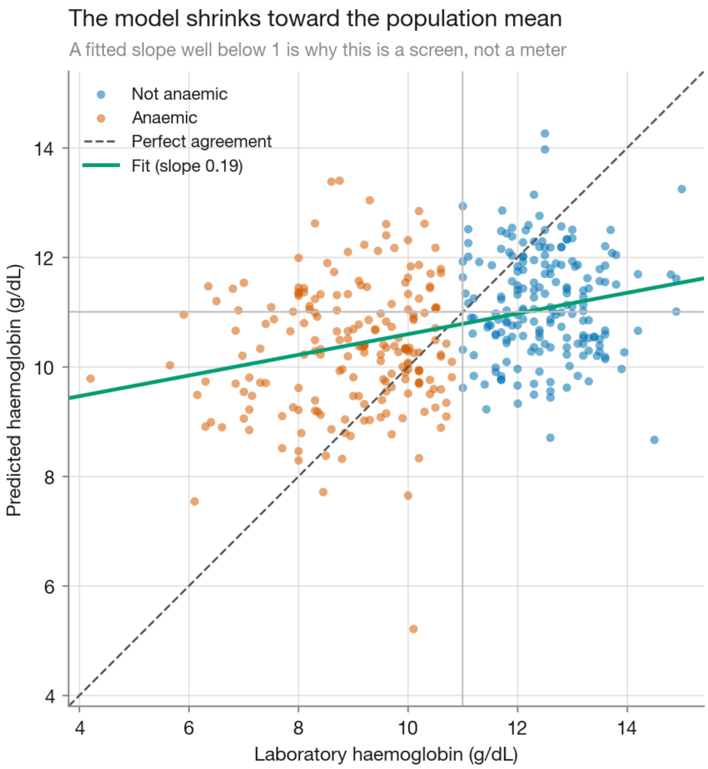

Figure 6: Predicted against laboratory haemoglobin. A least-squares fit through the predictions has slope 0.19, far below the identity line: a 1 g/dL change in true haemoglobin moves the prediction by roughly a fifth of that. The model separates anaemic from non-anaemic children while carrying little information about *how* anaemic they are.

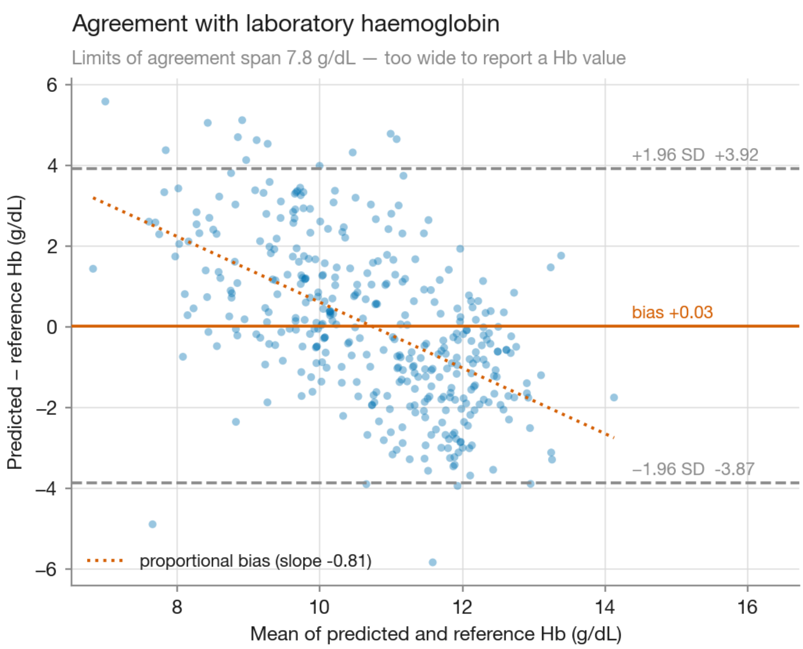

Figure 7: Bland–Altman agreement. The limits span 7.8 g/dL and the proportional bias is pronounced, establishing this as a triage screen rather than a measurement device.

4.7 Is acting on the model better than the alternatives?

Discrimination and calibration describe the model; neither says whether acting on it beats what a clinic would otherwise do. Where the realistic comparators are "refer every child" and "refer none", that comparison is what decides deployment. Decision curve analysis [4] makes it directly, through net benefit

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}}{N} - \frac{\mathrm{FP}}{N} \cdot \frac{p_t}{1-p_t}, \tag{1}$$

where p_t is the probability at which referral becomes worthwhile and the odds term prices one missed case against the false referrals a clinic will tolerate for it. A model is worth using only where its curve lies above both comparators.

Figure 8 shows the result, and it is a qualified one. **Below $p_t \approx 0.26$ the model is worse than simply referring every child.** That is a direct consequence of the 51 % prevalence in this cohort: when half the children are anaemic and missing a case is judged far costlier than an unnecessary referral, no triage rule beats referring everyone. Above that threshold the model does add benefit, and the margin widens as false referrals become more costly.

The practical reading is that this model earns its place only where confirmatory testing is scarce enough that indiscriminate referral is not an option. That is a narrower claim than AUROC 0.706 suggests on its own, and it is the claim the evidence supports. It is also a narrower window than the same analysis indicated before the third deduplication pass, when the crossing point stood at $p_t \approx 0.15$.

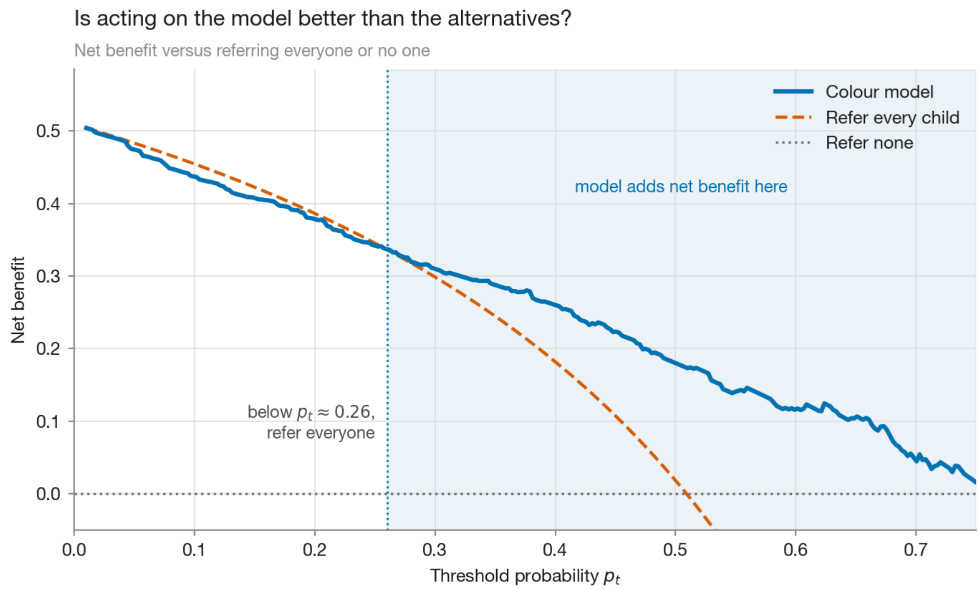

Figure 8: Decision curve analysis. Net benefit of the colour model against the two comparators available to a clinic. The model is not universally useful: below $p_t \approx 0.26$, referring every child is the better strategy, a consequence of the high prevalence in this cohort. Its advantage appears once false referrals carry real cost.

4.8 Cross-site generalisation

Leave-one-site-out evaluation across nine evaluable hospitals gives a mean AUROC of 0.722 (range 0.606–0.875); the tenth site contributes three photographs and cannot be scored alone. The highest score, 0.875 at SDA Hospital, rests on ten photographs at 80 % prevalence and should be read as noise rather than performance. The informative observation is that one of the two largest sites, Komfo Anokye Teaching Hospital ($n = 68$), returns the lowest score of all at 0.606, while the other, Ahmadiyya Muslim Hospital ($n = 68$), returns 0.747. Read the large sites rather than the mean: generalisation to an unseen hospital is nearer 0.7 than the 0.722 average, and the spread across sites is wide enough that a single-site pilot would not establish much.

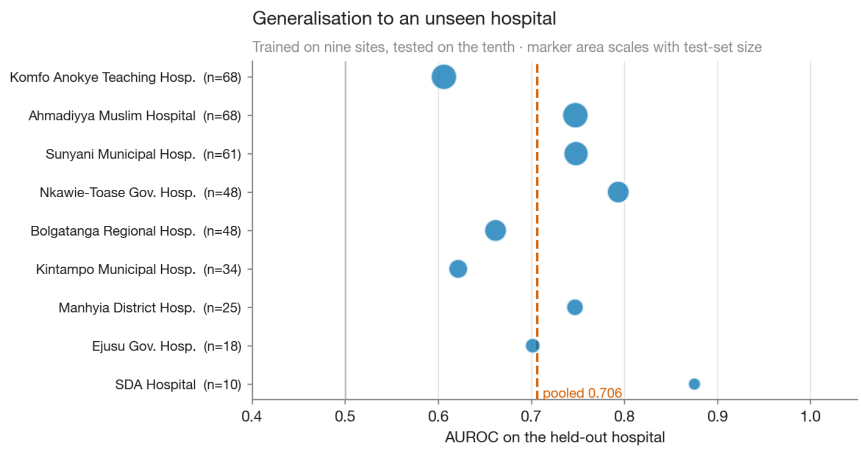

Figure 9: Leave-one-site-out performance. Marker area scales with test-set size; no interval is drawn, but the smallest sites (SDA, $n = 10$; Ejusu, $n = 18$) are estimated from too few children to support comparison, which is why we read the two largest sites instead.

4.9 Which features carry the signal

Permutation importance was measured out-of-fold (Figure 11), because the model's internal impurity importances are computed on training data and are biased toward continuous features. Four features share the lead within overlapping error bars: circular mean hue (ΔAUROC 0.042), mean saturation (0.041), the 10th percentile of the redness index (0.041) and the redness index itself (0.040), followed by mean red (0.036). Hue, saturation and the pale tail of the redness distribution are the physiologically expected carriers of pallor. Because these features are strongly correlated, a low score should be read as "redundant given the others", not "irrelevant". Figure 10 shows this rather than assuming it: mean absolute pairwise correlation across the twelve features is 0.52, reaching 0.98 between mean green and CIELAB L^*. Two clusters are visible — an RGB/lightness group and a saturation-and-redness group — and permutation importance divides credit within each. That is why no single feature exceeds 0.05 even though the set as a whole carries +0.182 AUROC over demographics.

4.10 Controls

Table 6 lists checks designed to make the headline result fail if it were an artefact. Three deserve comment.

The *byte-hash singles* control keeps only files whose SHA-256 appears exactly once. It scores 0.763, above the headline — and that is the expected direction, not a contradiction: re-encoded and shifted copies have unique hashes, so they survive this filter, and the subset is therefore not duplicate-free. We report it as a robustness check on the hash grain and no longer, as we did previously, as a subset "where no label conflict is possible".

Nested cross-validation now returns a *lower* score than the fixed hyperparameters (0.662 against 0.706). Tuning inside each fold at $n = 383$ overfits the inner split; the control's purpose — showing the fixed settings were not selected for flattery — is served, in the more convincing direction.

The *matched-n control* separates the two things pass 3 does. Subsampling the 479-photograph set to 383 rows without removing any duplicates scores 0.744 ± 0.021, midway between 0.780 and

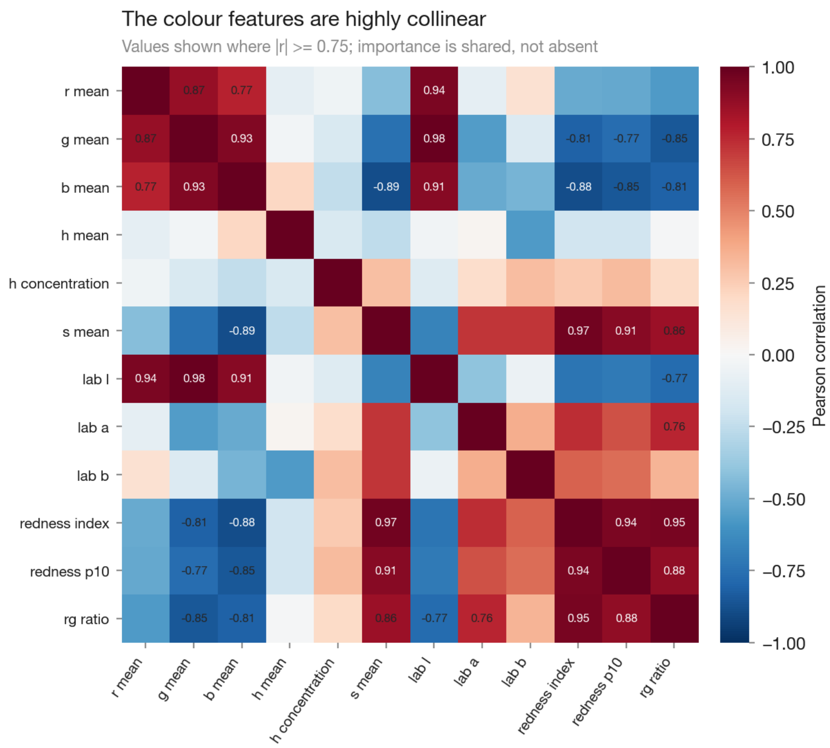

Figure 10: Pairwise correlation between the twelve colour features, printed where $|r| \geq 0.75$. This collinearity is why a low permutation-importance score should be read as redundancy given the rest of the set rather than absence of signal.

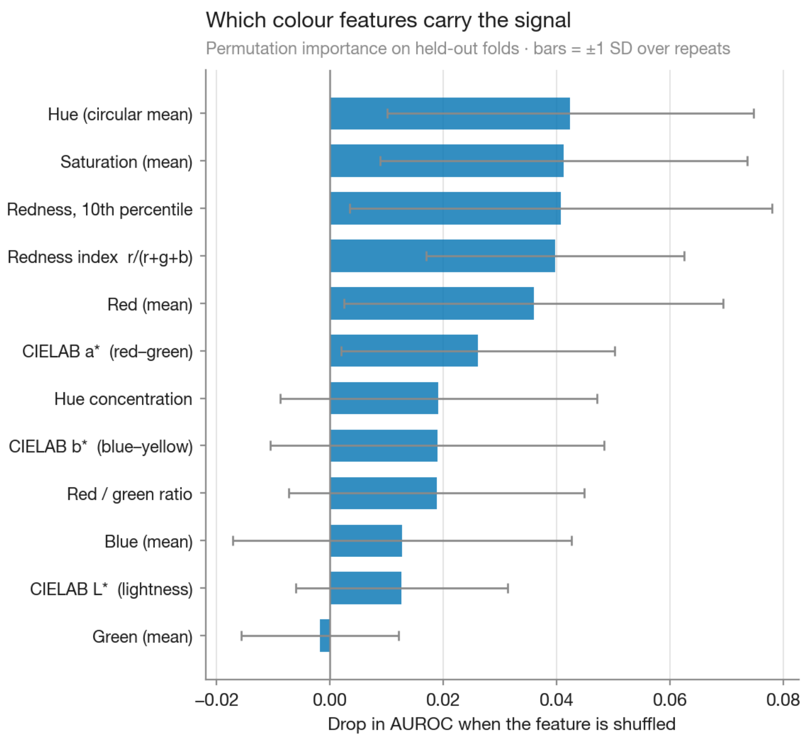

Figure 11: Out-of-fold permutation importance. Hue, saturation and the pale tail of the redness distribution lead, within overlapping error bars. Correlated features share credit, so a low score means redundant given the others rather than uninformative.

0.706: the pass-3 drop is about half sample size and about half leakage. This does not change the headline — 0.706 is what the model achieves on data that are actually distinct, which is the number a deployment would face — but it does bound how much of the fall is attributable to contamination.

The *alignment threshold sweep* is the least stable control. Between thresholds 1.0 and 4.0 the retained count falls from 397 to 376 while AUROC varies non-monotonically between 0.685 and 0.728: roughly ± 0.02 around the headline. The gap between merged and distinct pairs is real but narrower than at pass 2, so we report the range rather than a single number, and we chose the threshold by inspecting the borderline pairs by eye before seeing their effect on AUROC.

Table 6: Adversarial and robustness controls.

Control	Result	Interpretation
Label permutation (10 repeats)	AUROC 0.529 ± 0.032	chance; the harness does not leak
Seed stability (10 seeds)	0.702 ± 0.006	not fold-assignment luck
Byte-hash singles only ($n = 407$)	0.763	higher, as expected: copies survive this filter
Nested-CV hyperparameter tuning	$0.706 \rightarrow 0.662$	fixed settings not cherry-picked
Re-encode threshold sweep (0.5–5.0)	0.697–0.712	insensitive to the pass-2 cutoff
Alignment threshold sweep (1.0–4.0)	0.685–0.728, n 397–376	moderately sensitive; reported
Matched-n control for pass 3 (25 draws)	0.744 ± 0.021	half the pass-3 drop is sample size

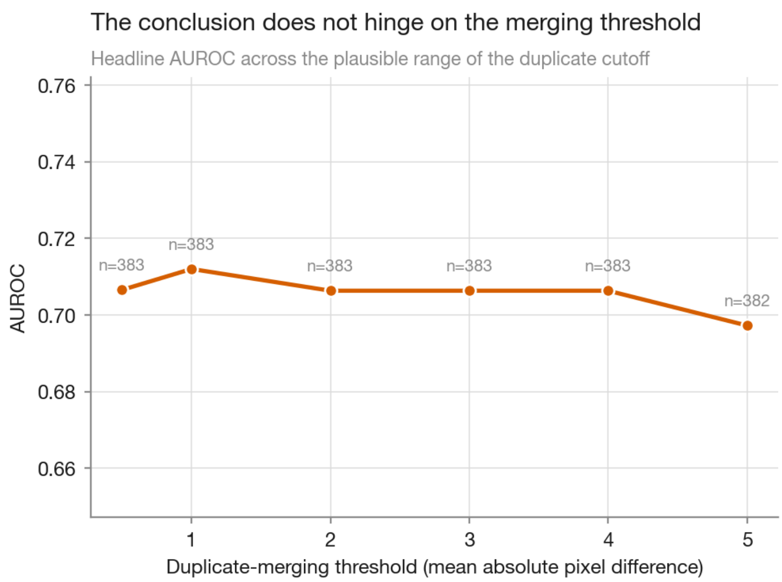

Figure 12: Sensitivity of the headline result to the pass-2 re-encode detection threshold. AUROC varies between 0.697 and 0.712 across the sweep, so the conclusion does not depend on where this cutoff is placed.

4.11 Would more data help?

A learning curve over training-set size (Figure 13) rises from AUROC 0.575 at a quarter of the data to 0.704 at full size and has not plateaued, though the ascent is not monotone — the 70 % point dips below the 55 % point, within the repeat-to-repeat spread the shaded band shows. Performance here is therefore plausibly limited by sample size as well as by the dataset's irreducible label noise — and the third deduplication pass has made the first constraint tighter, since the usable set is now 383 photographs rather than 479. A larger, cleanly labelled dataset would plausibly exceed 0.706.

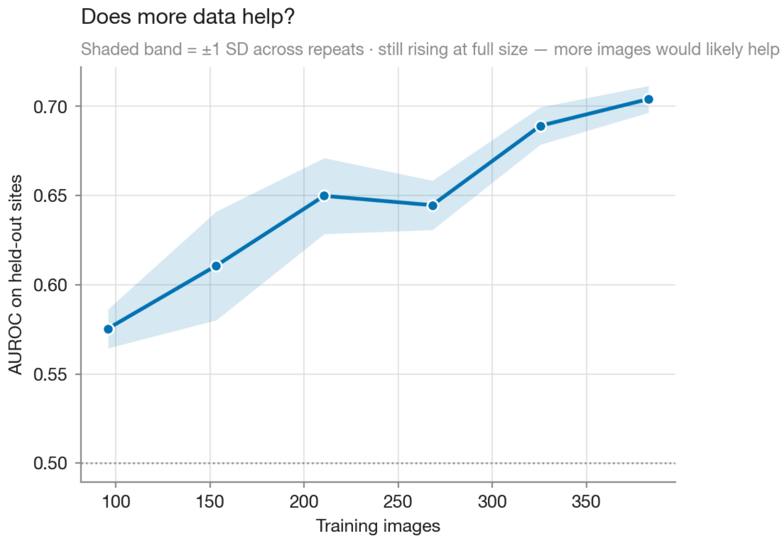

Figure 13: Learning curve. The absence of a plateau indicates the benchmark is sample-limited.

5 Discussion

The gap between 0.883 and 0.706 is an artefact of dataset handling rather than a modelling result. It matters because it is the kind of gap that separates a method reported as promising from one reported as marginal, and because nothing about the naive protocol looks careless from the inside — a random split is the default, and duplicates are invisible unless specifically sought.

Each form of duplication defeats the remedy for the previous one. The 19 re-encoded copies defeat a practitioner who deduplicates by file hash. The 96 shifted copies defeat the obvious repair: they survive pixel comparison too, because the hand-drawn mask moves with them, and 74 of the photographs they conceal are attributed to more than one hospital, so they survive site-grouped folds as well. Two independent safeguards, each individually sound, leave the leak in place.

Half of the last step's cost is sample size rather than contamination; the corrected number is lower both because the data are cleaner and because there are fewer of them, and a benchmark of 383 photographs is small for the task.

An earlier version of this analysis applied passes 1 and 2, grouped folds by site, and reported 0.780 as the corrected figure; the residual leak was found only when we searched specifically for copies whose masks disagreed. The general lesson we draw is that dataset audits should be reported as a specific list of what was tested for, since "deduplicated" names a procedure rather than a property, and each procedure has a blind spot that the next one exposes. We make no claim that a fourth form is absent from CP-AnemiC; we claim only that these three are present and that the detector for each is public.

Our corrected baseline should be read as establishing a floor for interpretable colour features, not a ceiling for the task. Learned representations may well exceed it. The purpose of reporting it is that any such claim can now be measured against a number produced under controls that are stated and reproducible.

Clinically, discrimination at AUROC 0.706 remains of some interest for triage where the alternative is no screening at all, but the margin has narrowed. The Bland–Altman result forecloses use as a haemoglobin estimator; the operating point — 70 % of healthy children referred at 90 % sensitivity — would impose a confirmatory testing burden that may exceed what a low-resource clinic can absorb; and the decision curve now places the useful window above $p_t \approx 0.26$ rather than 0.15. All three are more decision-relevant than the AUROC that such studies typically headline, and all three moved adversely when the evaluation set was corrected.

6 Limitations

- **Label noise is irreducible.** At the final grain, 116 groups carry conflicting haemoglobin on the same photograph; median merging cannot recover the truth, and a performance ceiling is therefore built into the benchmark.
- **Deduplication is a floor, not a guarantee.** We detect three forms of duplication and cannot exclude a fourth. The alignment pass searches integer translations only: a copy that was rotated, rescaled or re-illuminated would evade it. The retained count varies between 376 and 397 photographs across defensible thresholds (Section 4.10).
- **No colour calibration.** No colour card or white-balance reference was captured, so absolute RGB partly encodes illumination. Normalised ratios mitigate but do not eliminate this.
- **One country, one age band.** Children 6–59 months in Ghana. Nothing here supports use in adults, in pregnancy, or across different skin tones and camera pipelines.
- **Hand-drawn masks.** A deployed system would require automatic conjunctiva segmentation, whose error is not represented in any number reported here.
- **No true external validation.** Leave-one-site-out varies camera, operator and prevalence, but not country or protocol — and site attribution is itself unreliable here, since 125 duplicate groups span more than one hospital. Validation on an independent dataset is the obvious next step.
- **Not a clinical device.** No regulatory, ethical or safety assessment has been performed. This is a software study on a public dataset.

7 Conclusion

CP-AnemiC's 710 records resolve to 383 distinct photographs. The duplication takes three forms — byte-identical files, re-encoded copies, and shifted or re-cropped copies that survive both content hashing and aligned pixel comparison — and inflates AUROC by approximately 0.18 under a random split. Because 125 duplicate groups are attributed to more than one hospital, the duplication also defeats site-grouped cross-validation. We recommend that future work on this benchmark deduplicate under small translations, group folds by site, and state which forms of duplication were tested for.

Under those controls, twelve interpretable colour features achieve AUROC 0.706 (95 % CI 0.653–0.757) for WHO childhood anaemia, with the image contributing +0.182 over demographics alone. The resulting model is a plausible triage screen and is not a haemoglobin meter: limits of agreement span 7.8 g/dL and specificity falls to 0.303 at 90 % sensitivity. Reporting these quantities alongside AUROC is, we suggest, the minimum needed to judge whether such a method could be deployed.

Reproducibility

The complete analysis runs in approximately nine minutes on a laptop — dominated by the alignment pass; its pairwise distances are cached, and subsequent runs take about 90 seconds:

```
pip install -r requirements.txt
# download CP-AnemiC into data/cp-anemic/
python3 scripts/run_full_analysis.py
python3 -m pytest tests/ -q
```

Every number and figure in this paper is regenerated by that script. The accompanying test suite (62 tests) is organised by failure mode — split integrity, feature correctness, numerical robustness, determinism, statistical machinery, and adversarial nulls — on the principle that the dangerous failure in this setting is not a crash but a plausible wrong number.

Declarations

Use of generative AI. Manuscript text and analysis code were drafted with generative AI assistance (Anthropic Claude) under the author's direction. The author has verified the manuscript and takes full responsibility for its content.

Data availability

CP-AnemiC is publicly available from Mendeley Data (`m53vz6b7fx`) [1] and is used under its published licence. No patient-identifiable data is redistributed. Derived colour statistics only are stored in the accompanying repository.

Conflicts of interest

None declared.

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